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# Handbook of Modern Sensors

Physics, Designs, and Applications

Fourth Edition



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# Preface

Since publication of the previous, the 3rd edition of this book, the sensor technologies have made a remarkable leap ahead. The sensitivity of the sensors became higher, the dimensions – smaller, the selectivity – better, and the prices – lower. What have not changed, are the fundamental principles of the sensor design. They still are governed by the laws of Nature. Arguably one of the greatest geniuses ever lived, Leonardo Da Vinci had his own peculiar way of praying. It went like this, *“Oh Lord, thanks for Thou don’t violate Thy own laws.”* It is comforting indeed that the laws of Nature do not change with time, it is just that our appreciation of them becomes refined. Thus, this new edition examines the same good old laws of Nature that form the foundation for designs of various sensors. This has not changed much since the previous editions. Yet, the sections that describe practical designs are revised substantially. Recent ideas and developments have been added, while obsolete and less important designs were dropped.

This book is about devices commonly called *sensors*. The invention of a microprocessor has brought highly sophisticated instruments into our everyday life. Numerous computerized appliances, of which microprocessors are integral parts, wash clothes and prepare coffee, play music, guard homes, and control room temperature. Sensors are essential components in any device that uses a digital signal processor. The processor is a device that manipulates binary codes generally represented by electric signals. Yet, we live in an analog world, where such devices function among objects that are mostly not digital. Moreover, this world is generally not electrical (apart from the atomic level). Digital systems, however complex and intelligent they might be, must receive information from the outside world. Sensors are the interface devices between various physical values and electronic circuits that “understand” only a language of moving electrical charges. In other words, sensors are eyes, ears, and noses of silicon chips.

In the course of my engineering work, I often felt a strong need for a book which would combine practical information on diversified subjects related to the most important physical principles, design and use of various sensors. Surely, I could find almost all I had to know by surfing Internet or browsing library bookshelves in search for texts on physics, electronics, technical magazines, manufacturer’s

catalogues and websites. However, the information is scattered over many publications, and almost every question I was pondering required substantial research work. Little by little, I have been gathering practical information on everything, which in anyway was related to various sensors and their applications to scientific and engineering measurements. Soon, I realized that the information I collected might be quite useful to more than one person. This idea prompted me to write this book and this 4th edition is the proof that I was not mistaken.

In setting my criteria for selecting various sensors for the new edition, I attempted to keep the scope of this book as broad as possible, opting for many different designs described briefly (without being trivial, I hope), rather than fewer treated in greater depth. This volume attempts (immodestly perhaps) to cover a very broad range of sensors and detectors. Many of them are well known, but describing them is still useful for students and those who look for a convenient reference. It is the author's intention to present a comprehensive and up-to-date account of the theory (physical principles), design, and practical implementations of various (especially, the newest) sensors for scientific, industrial, and consumer applications. The topics included in the book reflect the author's own preferences and interpretations. Some may find a description of a particular sensor either too detailed or too broad or, on the contrary, too brief. In most cases, the author tried to strike a balance between a detailed description and simplicity of coverage.

It is clear that one book cannot embrace the whole variety of sensors and their applications, even if it would be called something like "*The Encyclopedia of Sensors*." This is a different book and the author's task was much less ambitious. Here, an attempt has been made to generate a reference text, which could be used by students, researchers interested in modern instrumentation (applied physicists and engineers), sensor designers, application engineers and technicians whose job is to understand, select and/or design sensors for practical systems.

The prior editions of this book have been used quite extensively as desktop references and textbooks for the related college courses. Comments and suggestions from the sensor designers, professors, and students prompted me to implement several changes and correct errors. I am deeply grateful to those who helped me to make further improvements in this new edition. I owe a debt of gratitude and many thanks to Drs. Ephraim Suhir and David Pintsov for assisting me in mathematical treatment of transfer functions and to Drs. Todd E. Mlsna and Sanjay V. Patel for their invaluable contribution to the chapter on chemical sensors.

Even though the book is intended for the scientific and engineering communities, as a rule, technical descriptions and mathematic treatments do not require a background beyond a high school curriculum. Simplicity of description and intuitive approach were the key requirements that I set for myself while working on the manuscript. My true goal was not to pile up a collection of information but rather to entice the reader into a creative process. As Plutarch said nearly two millennia ago, "*The mind is not a vessel to be filled but a fire to be kindled. . .*"

San Diego, California  
April, 2010

Jacob Fraden



# Chapter 1

## Data Acquisition

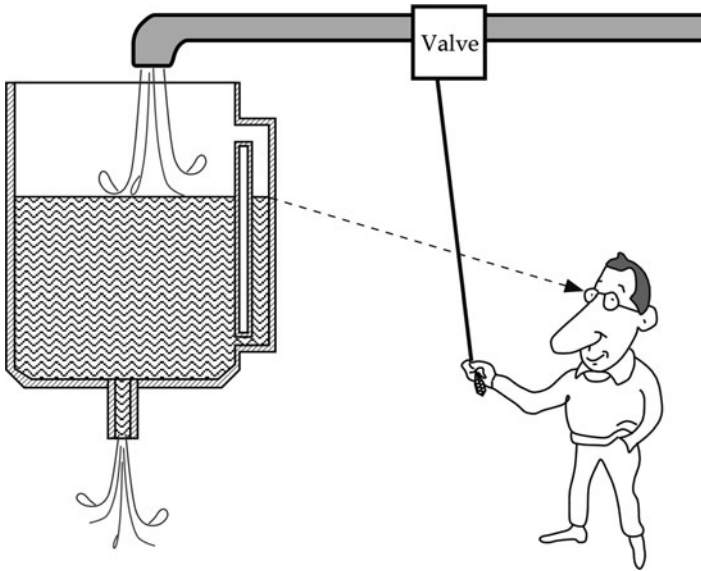
*It's as large as life, and twice as natural.*

– Lewis Carroll, *Through the Looking Glass*

### 1.1 Sensors, Signals, and Systems

A sensor is often defined as a “device that receives and responds to a signal or stimulus.” This definition is broad. In fact, it is so broad that it covers almost everything from a human eye to a trigger in a pistol. Consider the level-control system shown in Fig. 1.1 [1]. The operator adjusts the level of fluid in the tank by manipulating its valve. Variations in the inlet flow rate, temperature changes (these would alter the fluid’s viscosity and consequently the flow rate through the valve), and similar disturbances must be compensated for by the operator. Without control, the tank is likely to flood, or run dry. To act appropriately, the operator must obtain timely information about the level of fluid in the tank. In this example, the information is generated by the sensor, which consists of two main parts: the sight tube on the tank and the operator’s eye, which produces an electric response in the optic nerve. The sight tube by itself is not a sensor, and in this particular control system, the eye is not a sensor either. Only the combination of these two components makes a narrow-purpose sensor (detector), which is *selectively* sensitive to the fluid level. If a sight tube is designed properly, it will very quickly reflect variations in the level, and it is said that the sensor has a fast speed response. If the internal diameter of the tube is too small for a given fluid viscosity, the level in the tube may lag behind the level in the tank. Then, we have to consider a phase characteristic of such a sensor. In some cases, the lag may be quite acceptable, while in other cases, a better sight tube design would be required. Hence, the sensor’s performance must be assessed only as part of a data acquisition system.

This world is divided into natural and human-made objects. The natural sensors, like those found in living organisms, usually respond with signals, having an



**Fig. 1.1** Level control system. A sight tube and operator's eye form a sensor, a device which converts information into an electrical signal

electrochemical character, that is, their physical nature is based on ion transport, like in the nerve fibers (such as an optic nerve in the fluid tank operator). In man-made devices, information is also transmitted and processed in electrical form, however, through the transport of electrons. Sensors that are used in the artificial systems must speak the same language as the devices with which they are interfaced. This language is electrical in its nature and a man-made sensor should be capable of responding with signals where information is carried by displacement of electrons, rather than ions.<sup>1</sup> Thus, it should be possible to connect a sensor to an electronic system through electrical wires rather than through an electrochemical solution or a nerve fiber. Hence, in this book, we use a somewhat narrower definition of sensors, which may be phrased as

*A sensor is a device that receives a stimulus and responds with an electrical signal.*

The term *stimulus* is used throughout this book and needs to be clearly understood. The stimulus is the quantity, property, or condition that is received and converted into an electrical signal. Some texts (for instance, [2]) use a different term, *measurand* which has the same meaning, however with the stress on quantitative characteristic of sensing.

<sup>1</sup>There is a very exciting field of the optical computing and communications where information is processed by a transport of photons. That field is beyond the scope of this book.

The purpose of a sensor is to respond to some kind of an input physical property (stimulus) and to convert it into an electrical signal that is compatible with electronic circuits. We may say that a sensor is a translator of a generally nonelectrical value into an electrical value. When we say “electrical,” we mean a signal, which can be channeled, amplified, and modified by electronic devices. The sensor’s output signal may be in the form of voltage, current, or charge. These may be further described in terms of amplitude, polarity, frequency, phase, or digital code. This set of characteristics is called the *output signal format*. Therefore, a sensor has input properties (of any kind) and electrical output properties.

Any sensor is an energy converter. No matter what you try to measure, you always deal with energy transfer from the object of measurement to the sensor. The process of sensing is a particular case of information transfer, and any transmission of information requires transmission of energy. Of course, one should not be confused by an obvious fact that transmission of energy can flow both ways – it may be with a positive sign as well as with a negative sign; that is, energy can flow either from an object to the sensor or from the sensor to the object. A special case is when the net energy flow is zero, which also carries information about existence of that particular case. For example, a thermopile infrared radiation sensor will produce a positive voltage when the object is warmer than the sensor (infrared flux is flowing to the sensor) or the voltage is negative when the object is cooler than the sensor (infrared flux flows from the sensor to the object). When both the sensor and the object are at the same temperature, the flux is zero and the output voltage is zero. This carries a message that the temperatures are the same.

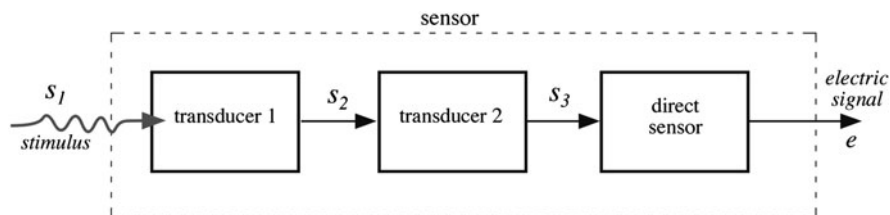
The term *sensor* should be distinguished from *transducer*. The latter is a converter of any one type of energy into another, whereas the former converts any type of energy into electrical energy. An example of a transducer is a loudspeaker, which converts an electrical signal into a variable magnetic field and, subsequently, into acoustic waves.<sup>2</sup> This is nothing to do with perception or sensing. Transducers may be used as *actuators* in various systems. An actuator may be described as an opposite to a sensor; it converts electrical signal into generally nonelectrical energy. For example, an electric motor is an actuator; it converts electric energy into mechanical action. Another example is a pneumatic actuator that is enabled by an electric signal.

Transducers may be parts of complex sensors (Fig. 1.2). For example, a chemical sensor may have a part, which converts the energy of a chemical reaction into heat (transducer) and another part, a thermopile, which converts heat into an electrical signal. The combination of the two makes a chemical sensor, a device which produces electrical signal in response to a chemical reagent. Note that in the above example a chemical sensor is a complex sensor; it is comprised of a nonelectrical transducer and a simple (direct) sensor converting heat to electricity. This suggests that many sensors incorporate at least one direct-type sensor and a

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<sup>2</sup>It is interesting to note that a loudspeaker, when connected to an input of an amplifier, may function as a microphone. In that case, it becomes an acoustical sensor.





**Fig. 1.2** A sensor may incorporate several transducers.  $s_1$ ,  $s_2$ , and so on are various types of energy. Note that the last part is a direct sensor producing electrical output  $e$

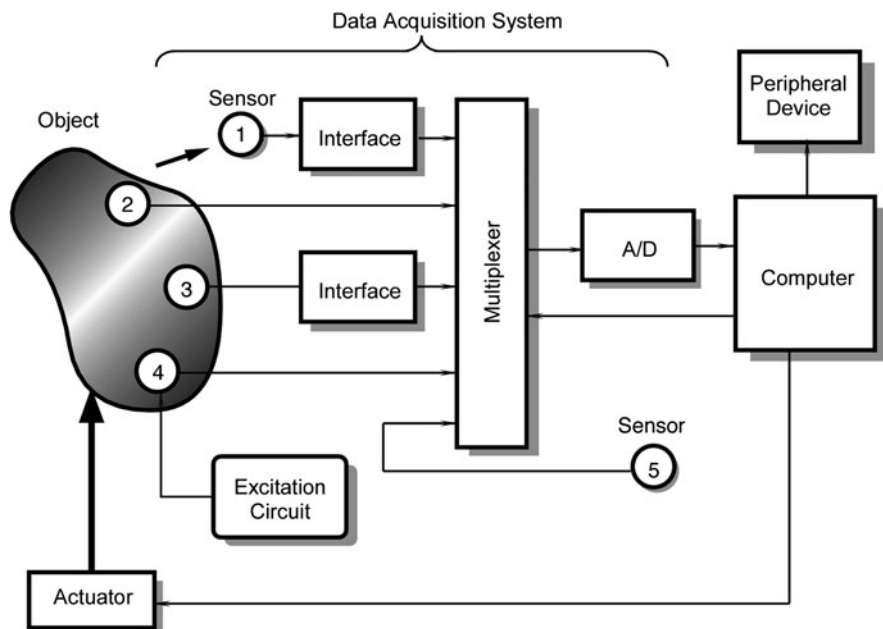
number of transducers. The direct sensors are those that employ certain physical effects to make a direct energy conversion into an electrical signal generation or modification. Examples of such physical effects are photoeffect and Seebeck effect. These will be described in Chap. 3.

In summary, there are two types of sensors; direct and complex. A direct sensor converts a stimulus into an electrical signal or modifies an electrical signal by using an appropriate physical effect, whereas a complex sensor in addition needs one or more transducers of energy before a direct sensor can be employed to generate an electrical output.

A sensor does not function by itself; it is always a part of a larger system that may incorporate many other detectors, signal conditioners, signal processors, memory devices, data recorders, and actuators. The sensor's place in a device is either intrinsic or extrinsic. It may be positioned at the input of a device to perceive the outside effects and to signal the system about variations in the outside stimuli. Also, it may be an internal part of a device that monitors the devices' own state to cause the appropriate performance. A sensor is always a part of some kind of a data acquisition system. Often, such a system may be a part of a larger control system that includes various feedback mechanisms.

To illustrate the place of sensors in a larger system, Fig. 1.3 shows a block diagram of a data acquisition and control device. An object can be anything: a car, space ship, animal or human, liquid, or gas. Any material object may become a subject of some kind of a measurement. Data are collected from an object by a number of sensors. Some of them (2, 3, and 4) are positioned directly on or inside the object. Sensor 1 perceives the object without a physical contact and, therefore, is called a noncontact sensor. Examples of such a sensor are a radiation detector and a TV camera. Even if we say "noncontact," we remember that energy transfer always occurs between any sensor and an object.

Sensor 5 serves a different purpose. It monitors internal conditions of a data acquisition system itself. Some sensors (1 and 3) cannot be directly connected to standard electronic circuits because of inappropriate output signal formats. They require the use of interface devices (signal conditioners). Sensors 1, 2, 3, and 5 are passive. They generate electric signals without energy consumption from the electronic circuits. Sensor 4 is active. It requires an operating signal, which is provided by an excitation circuit. This signal is modified by the sensor in



**Fig. 1.3** Positions of sensors in a data acquisition system. Sensor 1 is noncontact, sensors 2 and 3 are passive, sensor 4 is active, and sensor 5 is internal to a data acquisition system

accordance with the converted information. An example of an active sensor is a thermistor, which is a temperature-sensitive resistor. It needs a constant current source, which is an excitation circuit. Depending on the complexity of the system, the total number of sensors may vary from as little as one (a home thermostat) to many thousands (a space shuttle).

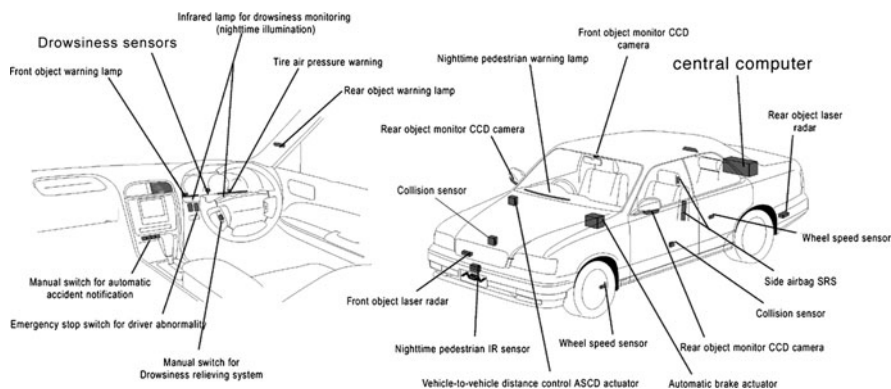
Electrical signals from the sensors are fed into a multiplexer (MUX), which is a switch or a gate. Its function is to connect sensors one at a time to an analog-to-digital converter (A/D or ADC) if a sensor produces an analog signal, or directly to a computer if a sensor produces signals in a digital format. The computer controls a multiplexer and an A/D converter for the appropriate timing. Also, it may send control signals to the actuator, which acts on the object. Examples of the actuators are an electric motor, a solenoid, a relay, and a pneumatic valve. The system contains some peripheral devices (for instance, a data recorder, a display, an alarm, etc.) and a number of components that are not shown in the block diagram. These may be filters, sample-and-hold circuits, amplifiers, and so forth.

To illustrate how such a system works, let us consider a simple car door monitoring arrangement. Every door in a car is supplied with a sensor, which detects the door position (open or closed). In most cars, the sensor is a simple electric switch. Signals from all door sensors go to the car's internal processor (no need for an A/D converter as all door signals are in a digital format: ones or

zeros). The processor identifies which door is open and sends an indicating signal to the peripheral devices (a dashboard display and an audible alarm). A car driver (the actuator) gets the message and acts on the object (closes the door).

An example of a more complex device is an anesthetic vapor delivery system. It is intended to control the level of anesthetic drugs delivered to a patient by means of inhalation during surgical procedures. The system employs several active and passive sensors. The vapor concentration of anesthetic agents (such as halothane, isoflurane, or enflurane) is selectively monitored by an active piezoelectric sensor installed into a ventilation tube. Molecules of anesthetic vapors add mass to the oscillating crystal in the sensor and change its natural frequency, which is a measure of vapor concentration. Several other sensors monitor the concentration of  $\text{CO}_2$  to distinguish exhale from inhale, and temperature and pressure, to compensate for additional variables. All of these data are multiplexed, digitized, and fed into the microprocessor, which calculates the actual vapor concentration. An anesthesiologist presets a desired delivery level and the processor adjusts the actuators (the valves) to maintain anesthetics at the correct concentration.

Another example of a complex combination of various sensors, actuators, and indicating signals is shown in Fig. 1.4. It is an advanced safety vehicle (ASV) that was developed by Nissan. The system is aimed at increasing safety of a car. Among many others, it includes a drowsiness warning system and drowsiness relieving system. This may include the eyeball movement sensor and the driver head inclination detector. The microwave, ultrasonic, and infrared range measuring sensors are incorporated into the emergency braking advanced advisory system to illuminate the break lamps even before the driver brakes hard in an emergency, thus advising the driver of a following vehicle to take evasive action. The obstacle warning system includes both the radar and infrared (IR) detectors. The adaptive cruise control system works like this: if the driver approaches too closely to a preceding vehicle, the speed is automatically reduced to maintain a suitable safety distance. The pedestrian monitoring system detects and alerts the driver to the



**Fig. 1.4** Multiple sensors, actuators, and warning signals are parts of the advanced safety vehicle (Courtesy of Nissan Motor Company)

presence of pedestrians at night as well as in vehicle blind spots. The lane control system helps in the event that the system detects and determines that incipient lane deviation is not the driver's intention. It issues a warning and automatically steers the vehicle, if necessary, to prevent it from leaving its lane.

In the following chapters we concentrate on methods of sensing, physical principles of sensors operations, practical designs, and interface electronic circuits. Other essential parts of the control and monitoring systems, such as actuators, displays, data recorders, data transmitters, and others are beyond the scope of this book and mentioned only briefly.

The sensor's input signals (stimuli) may have almost any conceivable physical or chemical nature (e.g., light, temperature, pressure, vibration, displacement, position, velocity, ion concentration, etc). The sensor's design may be of a general purpose. A special packaging and housing should be built to adapt it for a particular application. For instance, a micromachined piezoresistive pressure sensor may be housed into a water-tight enclosure for the invasive measurement of aortic blood pressure through a catheter. The same sensor will be given an entirely different enclosure when it is intended for measuring blood pressure by a noninvasive oscillometric method with an inflatable cuff. Some sensors are specifically designed to be very selective in a particular range of input stimulus and be quite immune to signals outside the desirable limits. For instance, a motion detector for a security system should be sensitive to movement of humans and not responsive to movement of smaller animals, like dogs and cats.

## 1.2 Sensor Classification

Sensor classification schemes range from very simple to the complex. Depending on the classification purpose, different classification criteria may be selected. Here, I offer several practical ways to look at the sensors.

1. All sensors may be of two kinds: passive and active. A passive sensor does not need any additional energy source and directly generates an electric signal in response to an external stimulus. That is, the input stimulus energy is converted by the sensor into the output signal. The examples are a thermocouple, a photodiode, and a piezoelectric sensor. Most of passive sensors are direct sensors as we defined them earlier.

The active sensors require external power for their operation, which is called an excitation signal. That signal is modified by the sensor to produce the output signal. The active sensors sometimes are called parametric because their own properties change in response to an external effect and these properties can be subsequently converted into electric signals. It can be stated that a sensor's parameter modulates the excitation signal and that modulation carries information of the measured value. For example, a thermistor is a temperature sensitive resistor. It does not generate

any electric signal, but by passing an electric current through it (excitation signal) its resistance can be measured by detecting variations in current and/or voltage across the thermistor. These variations (presented in ohms) directly relate to ttemperature through a known transfer function. Another example of an active sensor is a resistive strain gauge in which electrical resistance relates to a strain. To measure the resistance of a sensor, electric current must be applied to it from an external power source.

2. Depending on the selected reference, sensors can be classified into absolute and relative. An absolute sensor detects a stimulus in reference to an absolute physical scale that is independent of the measurement conditions, whereas a relative sensor produces a signal that relates to some special case. An example of an absolute sensor is a thermistor, a temperature-sensitive resistor. Its electrical resistance directly relates to the absolute temperature scale of Kelvin. Another very popular temperature sensor thermocouple is a relative sensor. It produces an electric voltage, which is a function of a temperature gradient across the thermocouple wires. Thus, a thermocouple output signal cannot be related to any particular temperature without referencing to a known baseline. Another example of the absolute and relative sensors is a pressure sensor. An absolute pressure sensor produces signal in reference to vacuum – an absolute zero on a pressure scale. A relative pressure sensor produces signal with respect to a selected baseline that is not zero pressure, for example, to the atmospheric pressure.
3. Another way to look at a sensor is to consider some of its properties that may be of a specific interest. Below are the lists of various sensor characteristics that may be considered (Tables 1.1–1.7).

**Table 1.1** Sensor specifications

| Sensitivity                      | Stimulus range (span)    |
|----------------------------------|--------------------------|
| Stability (short- and long-term) | Resolution               |
| Accuracy                         | Selectivity              |
| Speed of response                | Environmental conditions |
| Overload characteristics         | Linearity                |
| Hysteresis                       | Dead band                |
| Operating life                   | Output format            |
| Cost, size, weight               | Other                    |

**Table 1.2** Sensor material

| Inorganic            | Organic              |
|----------------------|----------------------|
| Conductor            | Insulator            |
| Semiconductor        | Liquid gas or plasma |
| Biological substance | Other                |

**Table 1.3** Detection means used in sensors

|                                            |
|--------------------------------------------|
| Biological                                 |
| Chemical                                   |
| Electric, magnetic or electromagnetic wave |
| Heat, temperature                          |
| Mechanical displacement or wave            |
| Radioactivity, radiation                   |
| Other                                      |

**Table 1.4** Conversion phenomena

|            |                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physical   | Thermoelectric<br>Photoelectric<br>Photomagnetic<br>Magnetoelectric<br>Electromagnetic<br>Thermoelastic<br>Electroelastic<br>Thermomagnetic<br>Thermooptic<br>Photoelastic<br>Other |
| Chemical   | Chemical transformation<br>Physical transformation<br>Electrochemical process Spectroscopy<br>Other                                                                                 |
| Biological | Biochemical transformation, Physical transformation<br>Effect on test organism Spectroscopy<br>Other                                                                                |

**Table 1.5** Field of applications

|                                       |                                    |
|---------------------------------------|------------------------------------|
| Agriculture                           | Automotive                         |
| Civil engineering, construction       | Domestic, appliances               |
| Distribution, commerce, finance       | Environment, meteorology, security |
| Energy, power                         | Information, telecommunication     |
| Health, medicine                      | Marine                             |
| Manufacturing                         | Recreation, toys                   |
| Military                              | Space                              |
| Scientific measurement                | Other                              |
| Transportation (excluding automotive) |                                    |

**Table 1.6** Stimulus

| Stimulus   |                                                           |
|------------|-----------------------------------------------------------|
| Acoustic   | Wave amplitude, phase, polarization                       |
|            | Spectrum                                                  |
|            | Wave velocity                                             |
|            | Other                                                     |
| Biological | Biomass (types, concentration, states)                    |
|            | Other                                                     |
| Chemical   | Components (identities, concentration, states)            |
|            | Other                                                     |
| Electric   | Charge, current                                           |
|            | Potential, voltage                                        |
|            | Electric field (amplitude, phase, polarization, spectrum) |
|            | Conductivity                                              |
|            | Permittivity                                              |
|            | Other                                                     |
| Magnetic   | Magnetic field (amplitude, phase, polarization, spectrum) |
|            | Magnetic flux                                             |
|            | Permeability                                              |
|            | Other                                                     |
| Optical    | Wave amplitude, phase, polarization, spectrum             |
|            | Wave velocity                                             |
|            | Refractive index                                          |
|            | Emissivity, reflectivity, absorption                      |
|            | Other                                                     |
| Mechanical | Position (linear, angular)                                |
|            | Acceleration                                              |
|            | Force                                                     |
|            | Stress, pressure                                          |
|            | Strain                                                    |
|            | Mass, density                                             |
|            | Moment, torque                                            |
|            | Speed of flow, rate of mass transport                     |
|            | Shape, roughness, orientation                             |
|            | Stiffness, compliance                                     |
|            | Viscosity                                                 |
|            | Crystallinity, structural integrity                       |
|            | Other                                                     |
| Radiation  | Type                                                      |
|            | Energy                                                    |
|            | Intensity                                                 |
|            | Other                                                     |
| Thermal    | Temperature                                               |
|            | Flux                                                      |
|            | Specific heat                                             |
|            | Thermal conductivity                                      |
|            | Other                                                     |

**Table 1.7** SI basic units

| Quantity                  | Name      | Symbol | Defined by... (year established)                                                                                                                                                             |
|---------------------------|-----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length                    | Meter     | m      | ...the length of the path traveled by light in vacuum in 1/299,792,458 of a second... (1983)                                                                                                 |
| Mass                      | Kilogram  | kg     | ...after a platinum-iridium prototype (1889)                                                                                                                                                 |
| Time                      | Second    | s      | ...the duration of 9,192,631,770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the cesium-133 atom (1967)                 |
| Electric current          | Ampere    | A      | Force equal to $2 \times 10^{-7}$ newton per meter of length exerted on two parallel conductors in vacuum when they carry the current (1946)                                                 |
| Thermodynamic temperature | Kelvin    | K      | The fraction 1/273.16 of the thermodynamic temperature of the triple point of water (1967)                                                                                                   |
| Amount of substance       | Mole      | mol    | ...the amount of substance which contains as many elementary entities as there are atoms in 0.012 kg of carbon 12 (1971)                                                                     |
| Luminous intensity        | Candela   | cd     | ...intensity in the perpendicular direction of a surface of 1/600,000 m <sup>2</sup> of a blackbody at temperature of freezing Pt under pressure of 101,325 newton per m <sup>2</sup> (1967) |
| Plane angle               | Radian    | rad    | (supplemental unit)                                                                                                                                                                          |
| Solid angle               | Steradian | sr     | (supplemental unit)                                                                                                                                                                          |

1.3 Units of Measurements

In this book, we use base units that have been established in The 14th General Conference on Weights and Measures (1971). The base measurement system is known as SI, which stands for Le Syst me International d’Unit s in French (Table 1.7) [4]. All other physical quantities are derivatives of these base units.<sup>3</sup> Some of them are listed in Table A-3.

Often it is not convenient to use base or derivative units directly; in practice quantities may be either too large or too small. For convenience in the engineering work, multiples and submultiples of the units are generally employed. They can be obtained by multiplying a unit by a factor from Table A-2 When pronounced, in all cases the first syllable is accented. For example, 1 ampere (A) may be multiplied by factor of 10<sup>−3</sup> to obtain a smaller unit 1 milliampere (mA), which is one thousandth of an ampere.

Sometimes, two other systems of units are used. They are the Gaussian System and the British System, which in the United States is modified as the U.S. Customary System. The United States is the only developed country where SI still is not in common use. However, with the increase of world integration, international cooperation gains a strong momentum. Hence, it is appears unavoidable that America will convert to SI in the future, though maybe not in our lifetime. Still, in

<sup>3</sup>The SI is often called the modernized metric system.



this book, we will generally use SI units, however, for the convenience of the reader, the U.S. customary system units will be used in places where U.S. manufacturers employ them for the sensor specifications. For the conversion to SI from other systems<sup>4</sup> the reader may use Tables A-4 of the Appendix. To make a conversion, a non-SI value should be multiplied by a number given in the table. For instance, to convert acceleration of 55 ft/s<sup>2</sup> to SI, it must to be multiplied by 0.3048:

$$55 \text{ ft/s}^2 \times 0.3048 = 16.764 \text{ m/s}^2$$

Similarly, to convert electric charge of 1.7 faraday, it must be multiplied by  $9.65 \times 10^{19}$ :

$$1.7 \text{ faraday} \times 9.65 \times 10^{19} = 1.64 \times 10^{20} \text{ C}$$

The reader should consider the correct terminology of the physical and technical terms. For example, in the United States and many other countries, electric potential difference is called “voltage,” while in other countries “electric tension” or simply “tension” is in common use (for example: *Spannung* in German and *напряжение* in Russian). In this book, we use terminology that is traditional in the United States of America.

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<sup>4</sup>Nomenclature, abbreviations, and spelling in the conversion tables are in accordance with ASTM SI10–02 IEEE/ASTM SI10 American National Standard for Use of the International System of Units (SI): The Modern Metric System. A copy is available from ASTM, 100 Barr Harbor Dr., West Conshocken, PA 19428–2959, USA. Tel.: (610) 832–9585, [www.astm.org/Standards/SI10.htm](http://www.astm.org/Standards/SI10.htm)